

QMWS Spring 2026: Introduction to Python for Data Analysis

Day One: Foundations

Gavin Klorfine & Deborah Laze

Section 1

Installation

- A tool for configuring Python **environments**
 - More often than not, you need to install **packages** that allow you to code more easily/efficiently
 - Allows for version control
 - Issues can arise if environments are not used, for example:
 - If you install two packages that conflict with one another
 - If you install a package that requires a specific version of Python
 - Anaconda allows these issues to be isolated to a specific environment, rather than affecting your whole Python installation
 - Especially useful if you have more than one project going on

- Installation link: <https://www.anaconda.com/download>
- The simplest way to use Anaconda is via the **Navigator**
- To install:
 - 1 Make an account
 - 2 Install the full distribution via the “Graphical Installer”

- Installation link: <https://www.anaconda.com/download>
- On Mac:



Choose Your Download

Windows

Mac

Linux

Anaconda Distribution

Complete package with 8,000+ libraries, Jupyter, JupyterLab, and Spyder IDE. Everything you need for data science.

↓ 64-Bit (Apple silicon)
Graphical Installer

↓ 64-Bit (Apple silicon)
Command Line Installer

Miniconda

Minimal installer with just Python, Conda, and essential dependencies. Install only what you need.

↓ 64-Bit (Apple silicon)
Graphical Installer

↓ 64-Bit (Apple silicon)
Command Line Installer

↓ 64-Bit (Intel chip) Graphical
Installer

↓ 64-Bit (Intel chip) Command

- Installation link: <https://www.anaconda.com/download>
- On Windows:



Choose Your Download

Windows

Mac

Linux

Anaconda Distribution

Complete package with 8,000+ libraries, Jupyter, JupyterLab, and Spyder IDE. Everything you need for data science.

[↓ Windows 64-Bit Graphical Installer](#)

Miniconda

Minimal installer with just Python, Conda, and essential dependencies. Install only what you need.

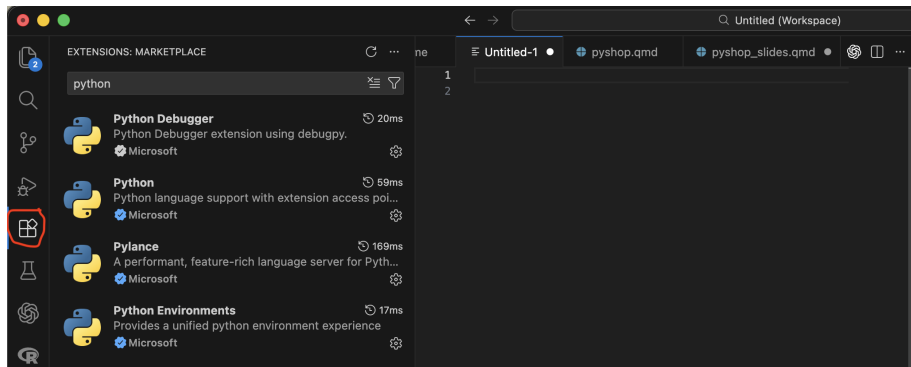
[↓ Windows 64-Bit Graphical Installer](#)

Visual Studio (VS) Code

- Installation link: <https://code.visualstudio.com/download>
- Integrated Development Environment (IDE)
- Can use it to code in many different languages
- Commonly used in both academia and industry

Visual Studio (VS) Code

- Once installed, go to the “Extensions” tab and search “Python”
- Install the below extensions:



Stop here!

If you:

- 1 Are able to open the **Anaconda Navigator**
- 2 Have installed **VS Code and the listed extensions**

You are ready for the workshop! We will cover next steps at the beginning of the first session.

Else, please email either of us for help:

- Gavin (gklorfin@yorku.ca)
- Bora (laze@yorku.ca)

Setting Up a Conda Environment

- See the presenter's shared screen
 - We will install the **ipykernel** package, allowing us to use **Jupyter notebooks** in VS Code (more on this in a few minutes)

- To point VS Code to your new Conda environment:
 - 1 Open the Command Palette
 - a. Ctrl + Shift + P on Windows
 - b. Command + Shift + P on MacOS
 - 2 Search “Python: Select Interpreter”
 - 3 Select the interpreter that corresponds to your new environment (the name of the environment will be in parentheses)

Scripts

- To create a new Python script (a file with a `.py` extension):
 - ① File -> New File -> Python File
- These scripts are generally executed from top to bottom, and are useful for writing longer chunks of code
- You can run a script with the play button in the top right corner of VS Code

Scripts

- To create a new Python script (a file with a `.py` extension):
 - 1 File -> New File -> Python File
- These scripts are generally executed from top to bottom, and are useful for writing longer chunks of code
- You can run a script with the play button in the top right corner of VS Code

Console

- The console is useful for executing smaller bits of code
- To open the console:
 - 1 Open the Command Palette -> search “Python: Start Terminal REPL”
- Type in a line of code and hit `enter` to execute it
 - Multiple lines may be typed by using `shift + enter`

Jupyter Notebooks

- Jupyter notebooks are a great way to combine code, text, and visualizations in one
- To create a new Jupyter notebook (a file with a `.ipynb` extension):
 - ① File -> New File -> Jupyter Notebook
- Notebooks are composed of **cells** that can be run independently of one another.
 - Cells can either be **code** or **markdown** (text)
- Click the play button beside a given cell to run it

Note

- If you would like to follow along by typing in the code from the slides, this may be easiest in a Jupyter notebook.

Section 2

The Basics

- In programming, a variable is a space in computer memory where values may be stored

Defining and Printing Variables

```
x = 67
```

```
print(x)
```

```
67
```

- Variable names should be short but descriptive (e.g., a variable for the mean of `x` should be named `mean_x`)
 - We are using `x` for pedagogical purposes... In practice, `x` is likely non-descriptive!

- Variable names should be short but descriptive (e.g., a variable for the mean of `x` should be named `mean_x`)
 - We are using `x` for pedagogical purposes... In practice, `x` is likely non-descriptive!
- When a variable name contains multiple words (e.g., apple size), there are a couple of approaches:
 - Use an underscore (e.g., `apple_size`)
 - Use “camel case” (e.g., `appleSize`)
 - Camel case has the first word all lowercase, with subsequent words having their first letter capitalized

Variables

- Variable names should be short but descriptive (e.g., a variable for the mean of x should be named `mean_x`)
 - We are using x for pedagogical purposes... In practice, x is likely non-descriptive!
- When a variable name contains multiple words (e.g., apple size), there are a couple of approaches:
 - Use an underscore (e.g., `apple_size`)
 - Use “camel case” (e.g., `appleSize`)
 - Camel case has the first word all lowercase, with subsequent words having their first letter capitalized
- If a variable is not meant to change, it is called a constant. These variables are usually named in all-caps, e.g.:

```
PI = 3.14
```

! Important

- Variable names cannot contain spaces (hence the use of camelCase and underscores)

! Important

- Variable names cannot contain spaces (hence the use of camelCase and underscores)
- Variable names cannot start with a number

! Important

- Variable names cannot contain spaces (hence the use of camelCase and underscores)
- Variable names cannot start with a number
- Avoid naming variables with names that have already been “taken”
 - E.g., do not name a variable “print”
 - We will encounter more names that have been “taken”

! Important

- Variable names cannot contain spaces (hence the use of camelCase and underscores)
- Variable names cannot start with a number
- Avoid naming variables with names that have already been “taken”
 - E.g., do not name a variable “print”
 - We will encounter more names that have been “taken”
- Do not capitalize the first letter of a variable unless it is a class (beyond the scope of this workshop)

Arithmetic

```
a = 1
```

```
b = 2
```

```
print(a + b)      # * for multiplication, / for division
```

```
3
```

Note

- Use # to make a **comment**

Updating Variables

```
x = 10
```

```
x = 20
```

```
x = 15
```

```
print(x)
```

```
15
```

Variable Types

- ❶ `int` (integer; “whole number”)
 - less memory
 - 1 is an `int`
- ❷ `float` (real; “decimal number”)
 - 1.2 is a `float`
 - 1.0 is a `float`
- ❸ `str` (string; characters/words with no numerical meaning)
 - “Hello world!” is a string
 - “416-676-6767” is also a string
- ❹ `bool` (boolean; can be `True` or `False`; more on this later)
- ❺ `None`
 - A special type of variable that essentially means “nothing meaningful is stored”
 - Useful as a placeholder

Variable Types

Checking Type

```
x = 416
print(x, type(x))    # type() to check type
416 <class 'int'>
```

Converting Type

```
x = 416
x = str(x)           # str() to convert to string
print(x, type(x))
```

```
416 <class 'str'>
```

- Can also use `int()` and `float()` to convert to respective type

A Brief Note

- `print()`, `type()`, `str()`, etc. are all examples of **functions**
 - These are essentially pre-made chunks of code for you to use
 - More on this later!

What will the below code output?

```
x = "Variable"

print(type(x))
print(x)
```

Section 3

More Advanced Variable Types

- Multiple values can be stored in one variable. A list is one way to do this:

Defining Lists

```
x = [0, 1, 2, 3, 4]      # Define a list; square brackets
print(type(x))
<class 'list'>
```

Lists: Indexing

```
x = [0,1,2,3,4]
```

- To extract one or more values from a list, you **index** the list
- Python uses **zero-based indexing**
 - This means that the first element of a list is index 0, the second element is index 1, ...
 - Element x is index $x - 1$

Indexing Lists

```
x = [0, 1, 2, 3, 4]
```

```
y = 2
```

```
print(x[0])      # Index first element of a list
```

```
print(x[-1])     # Index last element
```

```
print(x[y])      # Index yth element
```

```
0
```

```
4
```

```
2
```

Lists: Indexing

- You can also take more than one element, or a **slice**, of the list

Slicing Lists

```
x = [0, 1, 2, 3, 4]
```

```
print(x[0:2])    # First and second (but not third) elements  
[0, 1]
```

! Important

- [inclusive:exclusive]
 - The index on the *left-hand* side of the slice is *included*, whereas the index on the *right-hand* side of the slice is *excluded*

Slicing Lists, continued

```
x = [0, 1, 2, 3, 4]

print(x[0:3]) # First three elements
print(x[0:1]) # Silly way to index just the first element

[0, 1, 2]
[0]
```

Lists: Length

- To find the length of a list:

len() function

```
x = [0, 1, 2, 3, 4]
```

```
print(len(x))    # The length of list x
```

```
# Another way to find the last element (not so silly!)
```

```
print(x[len(x) - 1])
```

```
5
```

```
4
```

Lists: Methods

- Lists have certain functions that can be applied to them, called **methods**
 - Methods are more general than this (e.g., string methods; more on these in a little bit!)

`.append()`

```
x = [0,1,2,3,4]
```

```
x.append(5)
```

```
print(x)
```

```
[0, 1, 2, 3, 4, 5]
```

- Adds an entry to a list

`.extend()`

```
x = [0,1,2,3,4]
```

```
x.extend([6,7,8,9])
```

```
print(x)
```

```
[0, 1, 2, 3, 4, 6, 7, 8, 9]
```

- Extend a list by “glueing” it to another list

Difference Between `.append()` and `.extend()`

```
x = [0,1,2,4]
y = [5,6,7,8]
```

```
x.append(y)
print(x)
```

```
x = [0,1,2,4]    # Redefine x
x.extend(y)
print(x)
```

```
[0, 1, 2, 4, [5, 6, 7, 8]]
```

```
[0, 1, 2, 4, 5, 6, 7, 8]
```

Lists: Changing Values

Changing a Value in a List

```
x = [0, 1, 2, 3, 4]
```

```
x[0] = 9
```

```
print(x)
```

```
[9, 1, 2, 3, 4]
```

Tuples

- Like lists, tuples contain multiple values
- Unlike lists, tuples are **immutable**; they cannot be changed once created.
 - Lists are *mutable*

Defining a Tuple

```
x = ("a", "tuple") # Round brackets
```

```
print(x[0], x[1])
```

```
print(type(x))
```

```
a tuple
```

```
<class 'tuple'>
```

Tuples

Changing Values... ERROR!

```
x = ("a", "tuple")
```

```
x[0] = "Not a"
```

```
TypeError: 'tuple' object does not support item assignment
```

TypeError

Traceback (most recent call last):

Cell In[19], line 3

```
1 x = ("a", "tuple")
```

```
2
```

```
----> 3 x[0] = "Not a"
```

```
TypeError: 'tuple' object does not support item assignment
```

- Tuples have special properties, one of which is below:

Unpacking Tuples

```
x, y = (6, 7)
```

```
print(x)
```

```
print(y)
```

```
6
```

```
7
```

Section 4

Strings

- Strings are a lot like lists in that they are **iterable**
 - For now, think of these (lists, strings) as a sequence of values

Strings are Like Lists!

```
cringe = "Hello world!"
```

```
print(cringe[0:5])  
print(len(cringe))
```

```
Hello
```

```
12
```

- We can join two strings together through **concatenation**

Concatenation

```
a = "Hello"
```

```
b = "World!"
```

```
c = a + " " + b      # Note the space between words
```

```
print(c)
```

```
Hello World!
```

- Sometimes you need to include quotations or other characters inside a string that would normally return an error
- This can be done through an **escape sequence**

ERROR! Why You Need Escape Sequences...

```
x = "This is a "string"
```

```
SyntaxError: invalid syntax (886282512.py, line 1)
```

```
Cell In[23], line 1
```

```
    x = "This is a "string"  
                ^
```

```
SyntaxError: invalid syntax
```

Escape Sequences

```
x = "This is a \"string\""
print(x)
```

This is a "string"

Note

- The \ indicates the start of an escape sequence

Strings

- Strings are also kind of like tuples in that they are **immutable**

String Immutability

```
cringe = "Hello world!"  
cringe[0:5] = "Goodbye"      # Error
```

TypeError: 'str' object does not support item assignment

TypeError

Traceback (most recent call last):

Cell In[25], line 2

```
1 cringe = "Hello world!"
```

```
----> 2 cringe[0:5] = "Goodbye"      # Error
```

TypeError: 'str' object does not support item assignment

- We can turn to string **methods** to “modify” strings when needed (they just create a new string “behind the scenes”)

```
.replace()
```

```
cringe = "Hello world!"
```

```
print(cringe.replace("Hello", "Goodbye"))
```

```
Goodbye world!
```

`.split()`

```
meme_list = "Harambe, 67, Jar Jar Binks, The Rizzler"

# Split on ", "; returns a list
new_list = meme_list.split(", ")

print(new_list)
print(new_list[0]) # Index the list

['Harambe', '67', 'Jar Jar Binks', 'The Rizzler']
Harambe
```

How might you index “The Rizzler”?

```
memelist = "Harambe, 67, Jar Jar Binks, The Rizzler"  
new_list = memelist.split(", ")
```

How might you index “The Rizzler”?

```
meme_list = "Harambe, 67, Jar Jar Binks, The Rizzler"  
new_list = meme_list.split(", ")
```

```
print(new_list[3])  
print(new_list[-1])  
print(new_list[len(new_list) - 1])
```

The Rizzler

The Rizzler

The Rizzler

- If you wanted to include variables within text, format strings as per the following:

f-string Example

```
x = 3.14159
```

```
print(f'Approximate value of pi is {x}')
```

```
print(f'To three decimal places is {x:.3f}')
```

```
Approximate value of pi is 3.14159
```

```
To three decimal places is 3.142
```

- The string (surrounded by either single `' '` or double `" "` quotations) is preceded by `f`
- The variable is surrounded by curly braces `{}` within the string
- Formatting is applied to the variable *within the curly braces*
 - `A :` signals that the proceeding formatting code should be applied to the variable

f-string Formatting Basics

- `x:.3f`
 - Format variable `x`
 - Round to 3 decimal places
 - `f` means float

Section 5

Booleans

Booleans

- A boolean variable (type bool) can be either True (1) or False (0).

Booleans

```
x = True
y = 10
print(type(x), type(y))

print(y > 5)
print(y < 5)
print(type(y < 5))

<class 'bool'> <class 'int'>
True
False
<class 'bool'>
```

Booleans

```
x = 5
y = 10

print(x == 5)
print(x > 2 and y > 2)  # Both need to be True
print(x > 2 and y < 2)
print(x > 2 and not y < 2)
print(x > 2 or y < 2)  # At least one needs to be True
```

True

True

False

True

True

Comparison Operators

- > greater than
- < less than
- >= greater than or equal to
- <= less than or equal to
- == equal to
- != not equal to

Logical Operators

- and
- or
- not

Section 6

Conditional Statements

Conditional Statements

- If thing1 happens, do x
- If thing2 happens instead, do y
- Otherwise, do z

Conditional Statements

- If thing1 happens, do x
- If thing2 happens instead, do y
- Otherwise, do z

Basic if/elif/else

```
x = 10 # Value presumably unknown
y = 20
```

```
if x == 4:
    print("x is 4")
elif x > y:
    print("x is bigger than y")
else:
    print("none of the above")
```

```
none of the above
```

Conditional Statements

Structure of Conditionals

```
if x == 4:  
    print("x is 4")
```

- Begins with `if/elif` ("else if")/`else`
- Boolean expression/value
- Colon :
- Indented code underneath that runs if condition is met

Section 7

Loops

while Loops

- Loops are used to run a chunk of code repeatedly
 - Often a set number of times
- This is best shown and unpacked through examples:

while Loops

while Loop Basics

```
count = 1    # Initialize counter
```

```
while count <= 5:  
    print(count)  
    count = count + 1    # Increment counter
```

1

2

3

4

5

while Loops

Iterating through a list

```
x = [0,1,2,3]
count = 0

while count <= len(x) - 1:
    print(x[count])
    count += 1    # Same as count = count + 1
```

```
0
1
2
3
```

while Loops

- The `break` statement is used to exit a loop

`break`

```
x = 0

while True:      # Infinite loop...
    if x > 10:
        break    # Until you break!
    x += 1

print(x)
```

11

for Loops

- Often used with the `range()` function to loop over code a set number of times

for Loops, `range()`

```
for i in range(3):  
    print(i)
```

0

1

2

Iterating Over Entries in a List

```
x = [10,20,30,67]
```

```
for i in x:  
    print(i)
```

10

20

30

67

for Loops

- Use `enumerate()` and a second indexing variable (usually `i` or `j`) if you want to keep variable position

```
enumerate()
```

```
x = [10,20,30,67]
```

```
for i, j in enumerate(x):  
    print(i, j)
```

```
0 10
```

```
1 20
```

```
2 30
```

```
3 67
```

for Loops

- Loops (and conditionals) may be nested inside one another (like Russian dolls / Matryoshka)

Nesting

```
x = ["big", "small"]  
y = ["cat", "dog"]
```

```
for i in x:  
    for j in y:  
        print(i, j)
```

```
big cat  
big dog  
small cat  
small dog
```